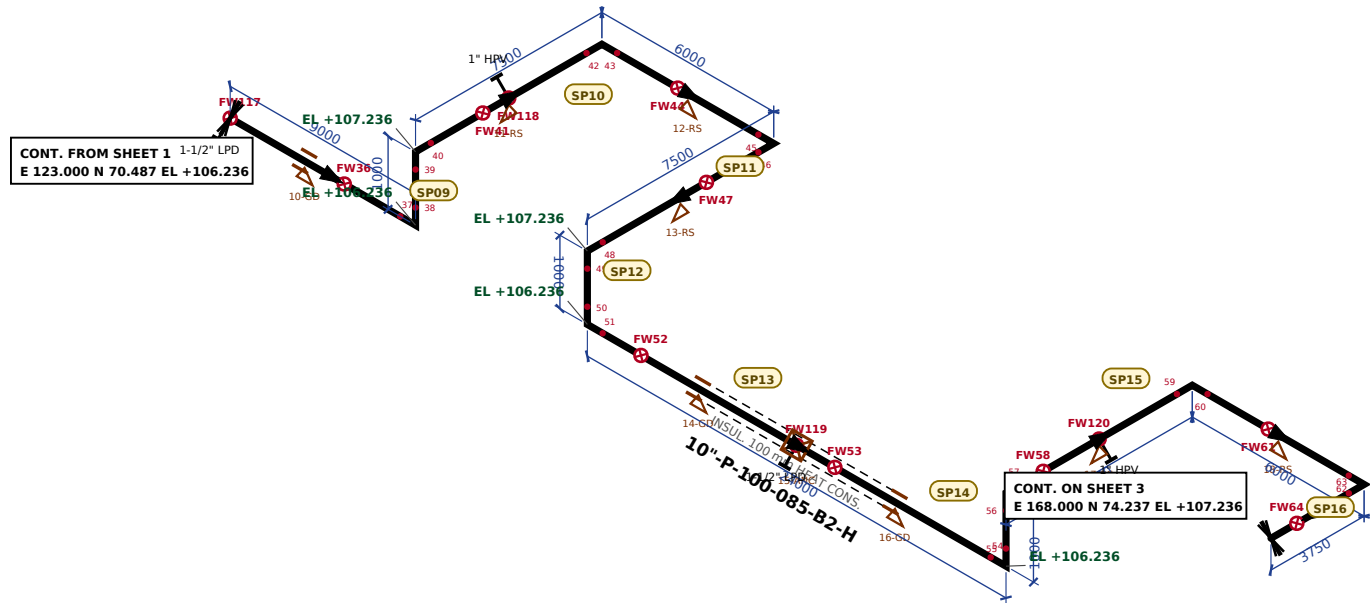
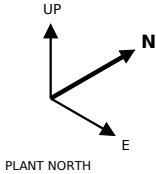


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE SCH 40, BE, ASTM A335 P5 SMLS	10"	60.2 m
2	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	10"	10
3	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	2
4	SOCKOLET 3000# (branch), ASTM A182 F5	1"	2
5	PIPE SUPPORTS (std. dwgs): anchor x1, guide x3, rest x5	-	9

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP09-1	10"	3288	SP09-2	10"	238	SP09-3	10"	2250
SP10-1	10"	4488	SP10-2	10"	2250	SP11-1	10"	2988
SP11-2	10"	2250	SP12-1	10"	4488	SP12-2	10"	238
SP12-3	10"	2250	SP13-1	10"	11700	SP14-1	10"	9288
SP14-2	10"	238	SP14-3	10"	950	SP15-1	10"	5788
SP15-2	10"	2250	SP16-1	10"	2988	SP16-2	10"	2250

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP09=7.3m SP10=7.5m SP11=6.0m SP12=8.5m SP13=11.7m SP14=12.0m SP15=8.8m SP16=6.0m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 44.7 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Discharge check + block valves in riser above each pump (accessible from grade).
- Rack run on tier 1 (hot B2) at rack south edge; expansion loops raised +1.0 m.
- FV-1083 / FE / XV station at grade (EL 101.2) north of H-201 for operator access.
- Riser at H-201 supported from heater structure with variable spring hangers.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 10"-P-100-085-B2-H P-112A/B DISCHARGE TO H-201 INLET		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 2 OF 3		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-004		REV A

LINE NUMBER 10"-P-100-085-B2-H	FROM P-112A/B	TO H-201 inlet manifold	PIPING CLASS B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T 24.0 barg / 380.0 C	OPER. P / T 18.99 barg / 348.7 C	TEST P (HYDRO) 44.7 barg (1.5xPxST/S=1.24)	PIPE WALL SCH 40 (9.27 mm)
INSULATION H 100 mm	PWHT YES - all thk (P-No.5B, B31.3 Tbl)	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; rotating equipment (
P&ID CFU-100-PR-PID-006, CFU-200-PR-PID-012	FLUID / PHASE P / L Atmospheric residue to H	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 33 spools; 89 shop / 32 field